K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

Lab Assignment-3

School of Engineering and Technology

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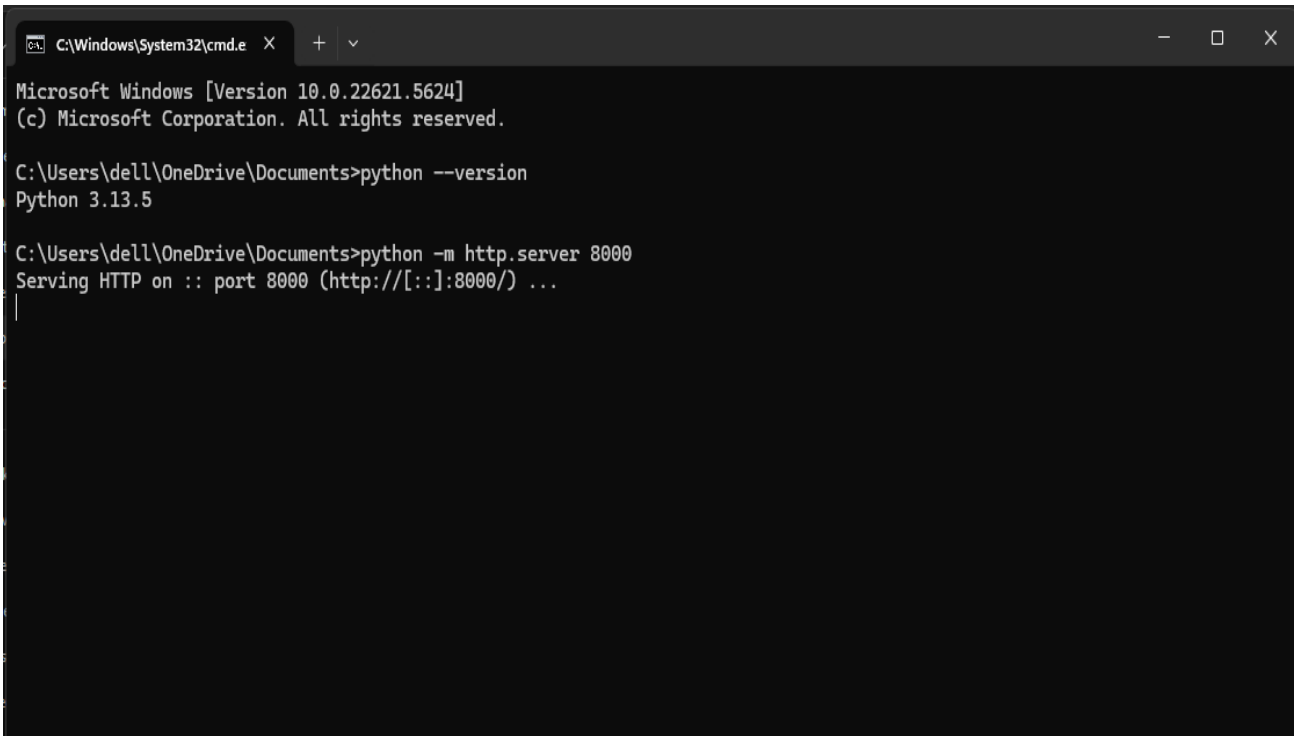
Program =BCA (AI & DS)

Big Data Analysis with Scala and spark Lab (ENSP359)

Submitted to :- Nikhil Sharma

Advertising Data Analytics using Cloudera & Spark-shell



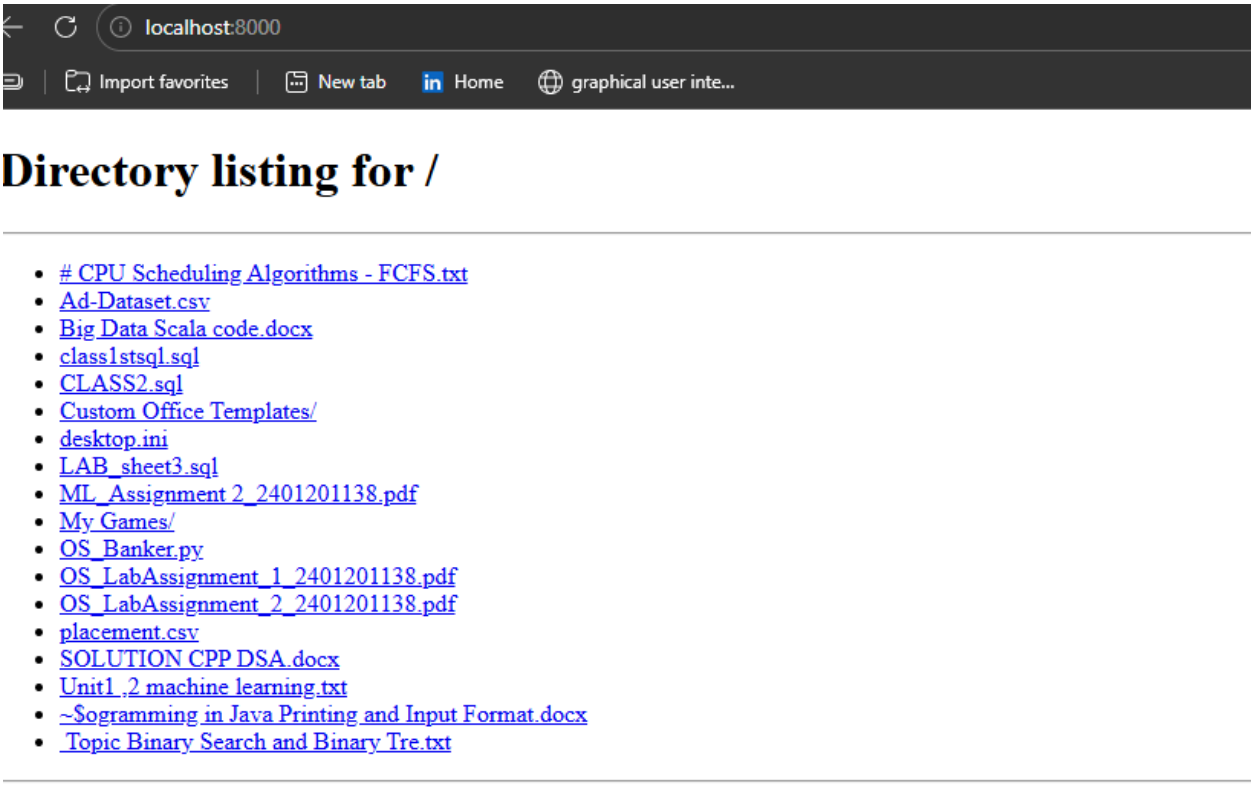
Task 1: Upload Dataset to Cloudera Environment

Commands Used:

1. `python --version` → This verifies Python is installed (yours shows Python 3.13.5)
2. `python -m http.server 8000` → This starts a local HTTP server on port 8000 so your Cloudera VM can download the dataset file from your Windows machine over the network

Output: → `Serving HTTP on :: port 8000 (http://[::]:8000/) ...` — confirms the server is running

Purpose: → By running this HTTP server, the Cloudera environment can access your local files and you can upload the Advertising dataset to HDFS



Verification — HTTP Server Working:

→ Opened browser and went to `http://localhost:8000`

→ It shows **Directory listing for** — all files from your Documents folder are visible

→ You can see **`Ad-Dataset.csv`** in the list — this is the Advertising dataset that will be uploaded to Cloudera/HDFS

→ The Cloudera VM can now access this file using: `wget http://<your-windows-IP>:8000/Ad-Dataset.csv`

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22621.5624]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::318b:4d8d:2160:33a0%17
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : bbrouter
   Link-local IPv6 Address . . . . . : fe80::3258:4311:5037:3ad3%27
   IPv4 Address. . . . . . . . . . . : 192.168.1.42
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%27
                                       192.168.1.1

Mobile Broadband adapter Cellular:
```

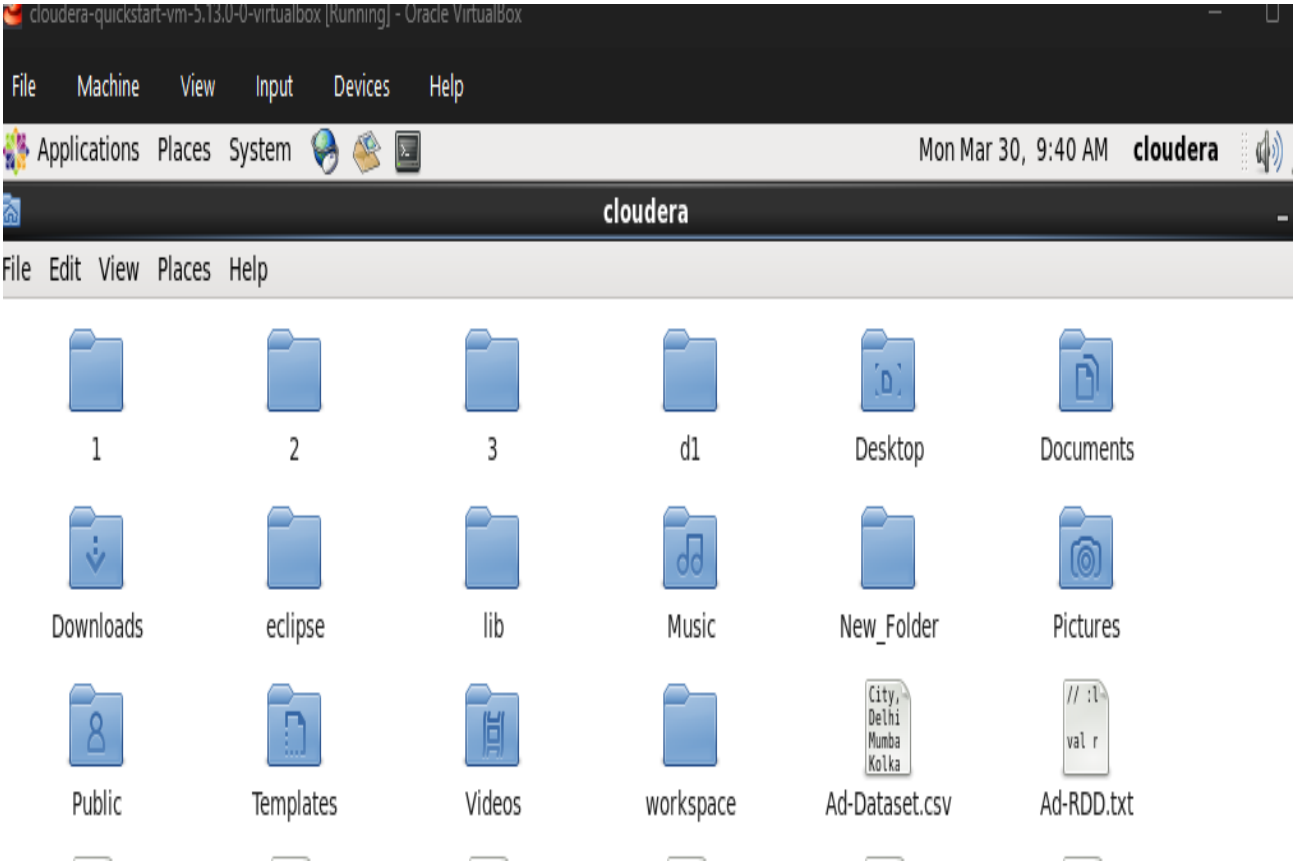
Finding Windows IP Address:

→ Command used: `ipconfig`

→ **Wi-Fi IPv4 Address:** **`192.168.1.42`** ← this is the important one

→ This IP is needed so the Cloudera VM can connect to your Windows HTTP server

→ The Cloudera VM would use this command to download the dataset:
`wget http://192.168.1.42:8000/Ad-Dataset.csv`



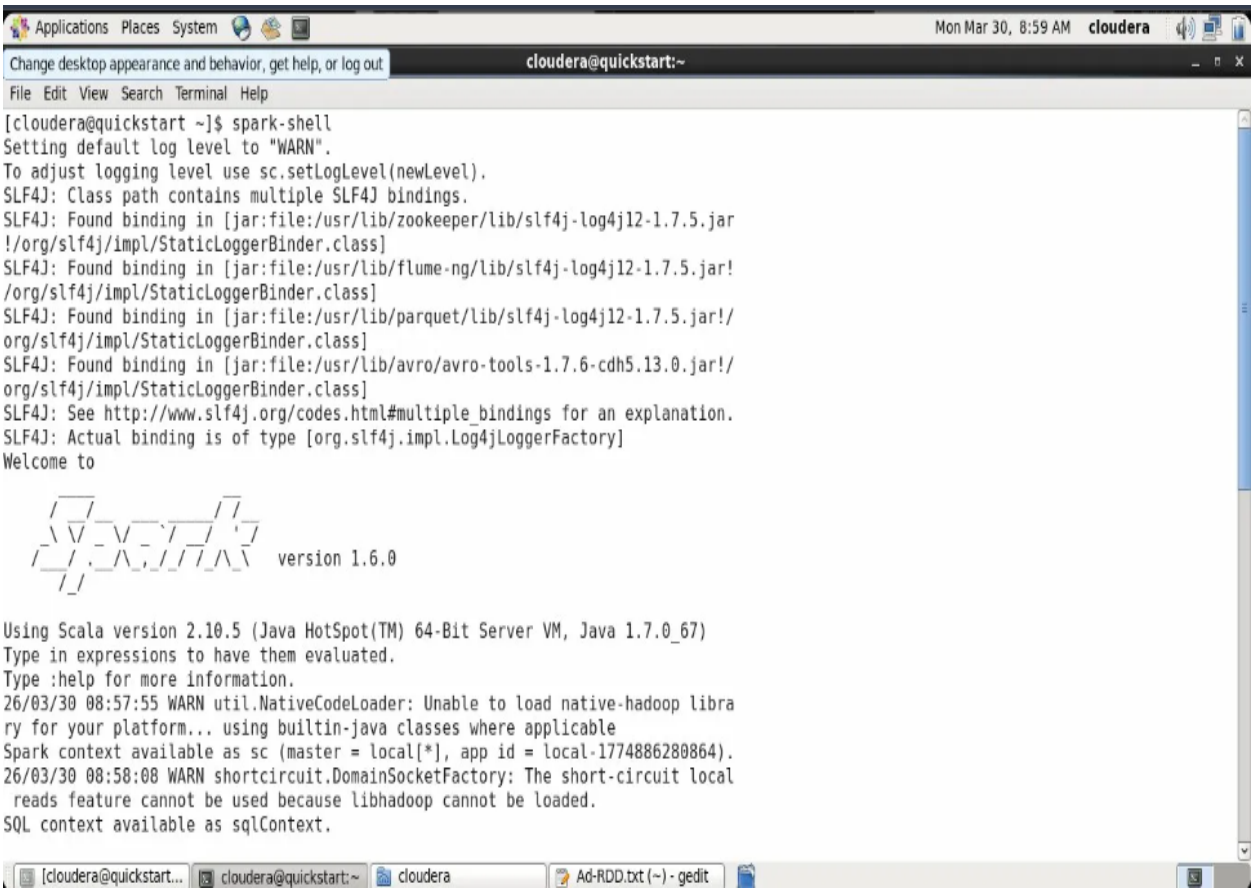
Dataset Available in Cloudera VM:

→ This shows the **Cloudera QuickStart VM** running in Oracle VirtualBox

→ You can see `Ad-Dataset.csv` file is present in the Cloudera file system — successfully transferred!

→ You can also see `Ad-RDD.txt` file — this is the query file required for Task 3

→ This confirms the dataset was successfully downloaded from the Windows HTTP server into the Cloudera environment



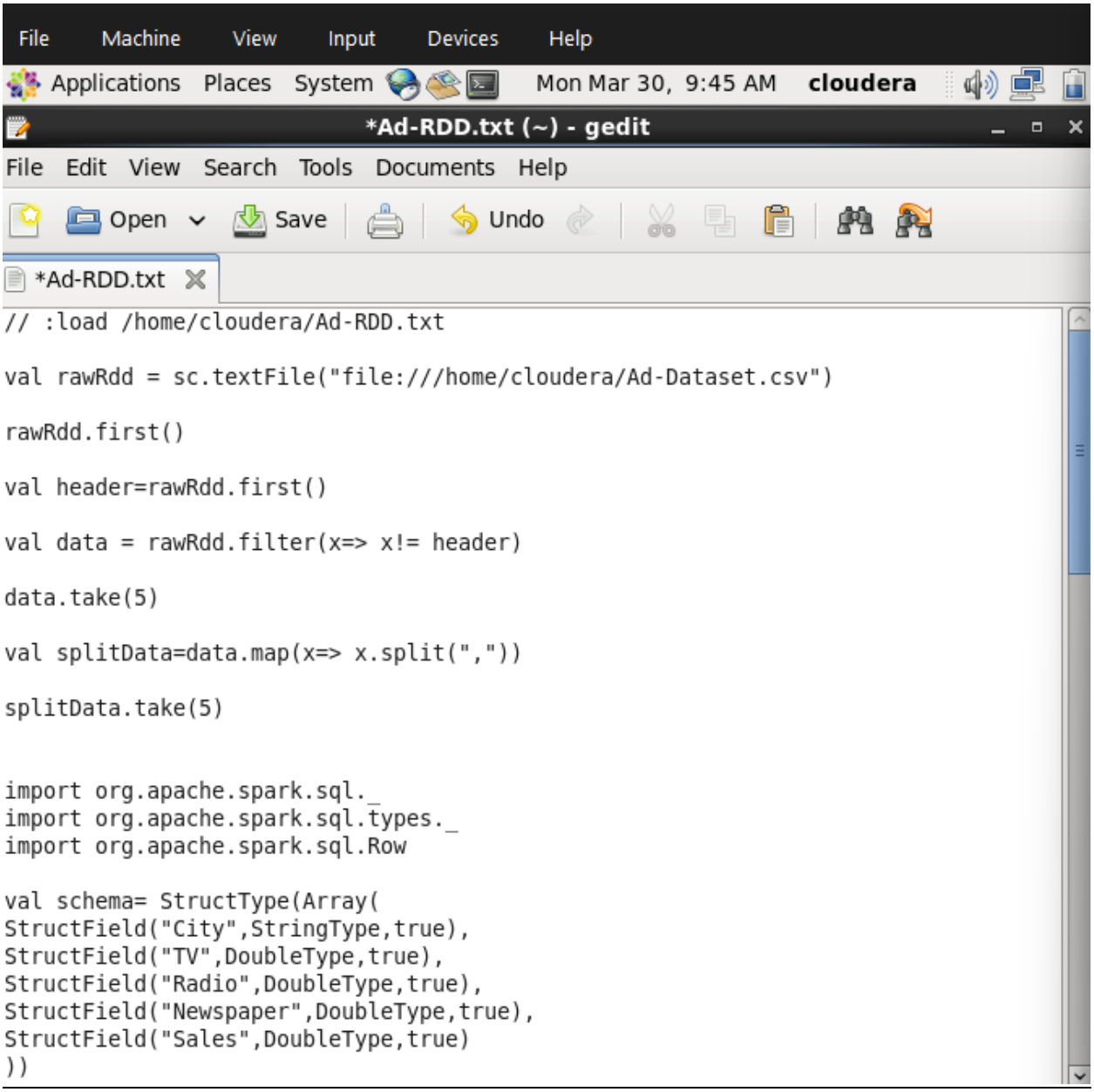
Starting Spark-Shell:

→ Command used: `spark-shell`

→ Spark shell launched successfully with **Scala version 2.10.5** and **Java 1.7.0_67**

→ **Spark context available as `sc`** (master = local, app id = local-177488628864)

→ **SQL context available as `sqlContext`**




```
// :load /home/cloudera/Ad-RDD.txt

val rawRdd = sc.textFile("file:///home/cloudera/Ad-Dataset.csv")

rawRdd.first()

val header=rawRdd.first()

val data = rawRdd.filter(x=> x!= header)

data.take(5)

val splitData=data.map(x=> x.split(","))

splitData.take(5)


import org.apache.spark.sql._
import org.apache.spark.sql.types._
import org.apache.spark.sql.Row

val schema= StructType(Array(
StructField("City",StringType,true),
StructField("TV",DoubleType,true),
StructField("Radio",DoubleType,true),
StructField("Newspaper",DoubleType,true),
StructField("Sales",DoubleType,true)
))
```

Task2: Spark Processing Code (Ad-RDD.txt):

Step 1 — Load dataset as RDD:

→ Loads the CSV file using **SparkContext (`sc`)** into an RDD called `rawRdd` → `.first()` shows the first row (header)

Step 2 — Remove Header:

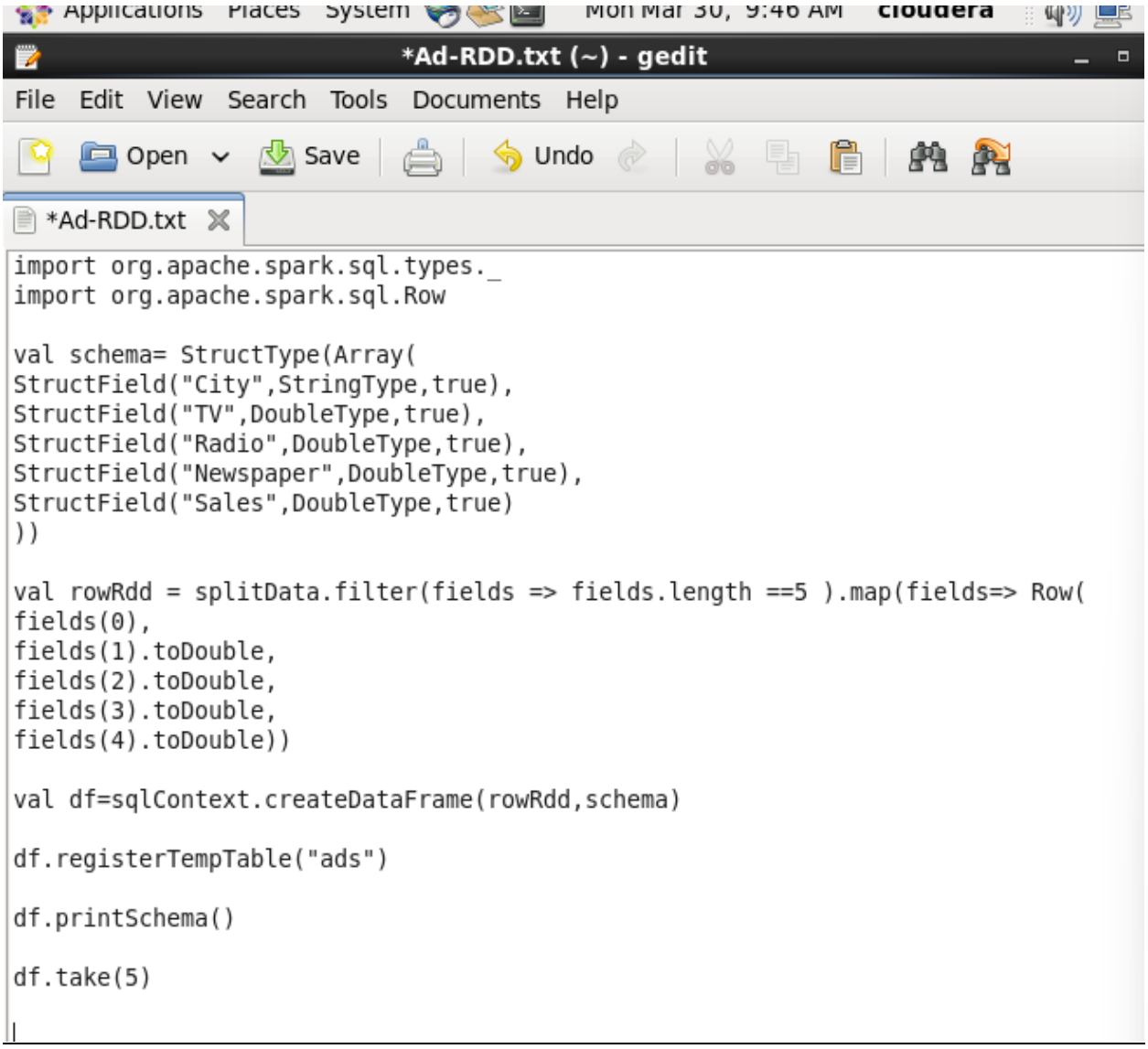
→ Saves the header row, then filters it out from the data

Step 3 — Split each row by comma:

Each row is split into an array of values

Step 4 — Define Schema:

Defines column names and data types for the DataFrame




```
import org.apache.spark.sql.types._
import org.apache.spark.sql.Row

val schema= StructType(Array(
StructField("City",StringType,true),
StructField("TV",DoubleType,true),
StructField("Radio",DoubleType,true),
StructField("Newspaper",DoubleType,true),
StructField("Sales",DoubleType,true)
))

val rowRdd = splitData.filter(fields => fields.length ==5 ).map(fields=> Row(
fields(0),
fields(1).toDouble,
fields(2).toDouble,
fields(3).toDouble,
fields(4).toDouble))

val df=sqlContext.createDataFrame(rowRdd,schema)

df.registerTempTable("ads")

df.printSchema()

df.take(5)
```

Step 5 — Convert RDD to DataFrame:

Filters only rows with exactly 5 fields (removes bad/empty rows) → Maps each row into a `Row` object with correct data types

Step 6 — Create DataFrame:

→ Combines the `rowRdd` and `schema` to create a proper DataFrame

Step 7 — Register as Table:

Registers the DataFrame as a **temporary SQL table named `ads`** so SQL queries can be run on it

Step 8 — Verify:

`printSchema()` shows column names and types → `take(5)` shows first 5 records to confirm data loaded correctly

Task:-3 ALL Queries

```
//1st Total number of records
df.count()


//2nd Total and avergae sales
df.agg(sum("Sales").as("Total_Sales"),avg("Sales").as("Average_Sales")).show()


//3) City-wise Total Sales
df.groupBy("City").sum("Sales").show()


//4  City-wise Average Sales sorted Descending
df.groupBy("City").agg(avg("Sales").as("Average_Sales")).orderBy(desc("Average_Sales")).show()


//5) City wise Maximum Sales
df.groupBy("City").agg(max("Sales").as("Max_Sales")).show()


//6) Number of records in each city sorted by count
df.groupBy("City").count().orderBy(desc("count")).show()


//7) Total TV spend per city where total spend >500
df.groupBy("City").agg(sum("TV").as("Total_TV")).filter(col("Total_TV")>500).show()
```

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cloudera@quicksta... | cloudera@quicksta... | cloudera | *Ad-RDD.txt (~) - g...

cloudera-quickstart-vm-5.13.0-0-virtualbox [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Applications Places System Mon Mar 30, 9:48 AM cloudera

*Ad-RDD.txt (~) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

*Ad-RDD.txt

```
//6) Number of records in each city sorted by count
df.groupBy("City").count().orderBy(desc("count")).show()


//7) Total TV spend per city where total spend >500
df.groupBy("City").agg(sum("TV").as("Total_TV")).filter(col("Total_TV")>500).show()


//8)Average radio spend per city where avg> 20
df.groupBy("City").agg(avg("Radio").as("Avg_Radio")).filter(col("Avg_Radio") > 20).show()


//9) Top 5 cities with highest average sales
df.groupBy("City").agg(avg("Sales").as("Avg_Sales")).orderBy(desc("Avg_Sales")).limit(5).show()


//10 Cities where average sales is between 10 and 20
df.groupBy("City").agg(avg("Sales").as("Avg_Sales")).filter(col("Avg_Sales").between(10,20)).show()


//11) Records where TV > 200 ans sales > 15
df.filter (col("TV") > 200 && col("Sales") > 15).show()


//12 Records where Radio < 10 Or Newspaper > 50
df.filter (col("Radio") < 10 || col("Newspaper") > 50).show()
```

```
scala> :load /home/cloudera/Ad-RDD.txt
Loading /home/cloudera/Ad-RDD.txt...
rawRdd: org.apache.spark.rdd.RDD[String] = file:///home/cloudera/Ad-Dataset.csv M
apPartitionsRDD[88] at textFile at <console>:34
res20: String = City,TV,Radio,Newspaper,Sales
header: String = City,TV,Radio,Newspaper,Sales
data: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[89] at filter at <conso
le>:38
res21: Array[String] = Array(Delhi,230.1,37.8,69.2,22.1, Mumbai,44.5,39.3,45.1,10
.4, Kolkata ,17.2,45.9,69.3,12, Mumbai,151.5,41.3,58.5,16.5, Delhi,180.8,10.8,58.
4,17.9)
splitData: org.apache.spark.rdd.RDD[Array[String]] = MapPartitionsRDD[90] at map
at <console>:40
res22: Array[Array[String]] = Array(Array(Delhi, 230.1, 37.8, 69.2, 22.1), Array(
Mumbai, 44.5, 39.3, 45.1, 10.4), Array("Kolkata ", 17.2, 45.9, 69.3, 12), Array(M
umbai, 151.5, 41.3, 58.5, 16.5), Array(Delhi, 180.8, 10.8, 58.4, 17.9))
import org.apache.spark.sql._
import org.apache.spark.sql.types._
import org.apache.spark.sql.Row
schema: org.apache.spark.sql.types.StructType = StructType(StructField(City,Strin
gType,true), StructField(TV,DoubleType,true), StructField(Radio,DoubleType,true),
 StructField(Newspaper,DoubleType,true), StructField(Sales,DoubleType,true))
rowRdd: org.apache.spark.rdd.RDD[org.apache.spark.sql.Row] = MapPartitionsRDD[92]
 at map at <console>:49
```

Loading the file: → `:load /home/cloudera/Ad-RDD.txt` — file loaded successfully

RDD created: → `rawRdd` = RDD with 34 partitions, dataset loaded from `Ad-Dataset.csv`

Header extracted: → `header: String = City,TV,Radio,Newspaper,Sales`

Data after removing header: → Shows actual data rows like: `Delhi, 230.1, 37.8, 69.2, 22.1` etc. — 38 records

splitData created: → Each row split into array of values — 40 partitions

Schema defined successfully: → `StructType` with fields: City(String), TV(Double), Radio(Double), Newspaper(Double), Sales(Double)

rowRdd created: → RDD of Row objects — 49 partitions, ready to become DataFrame

This shows DataFrame created + Query 1 & 2 & 3 outputs

```
import org.apache.spark.sql._
import org.apache.spark.sql.types._
import org.apache.spark.sql.Row
schema: org.apache.spark.sql.types.StructType = StructType(StructField(City,StringType,true), StructField(TV,Dou
bleType,true), StructField(Radio,DoubleType,true), StructField(Newspaper,DoubleType,true), StructField(Sales,Dou
bleType,true))
rowRdd: org.apache.spark.rdd.RDD[org.apache.spark.sql.Row] = MapPartitionsRDD[100] at map at <console>:56
df: org.apache.spark.sql.DataFrame = [City: string, TV: double, Radio: double, Newspaper: double, Sales: double]
root
 |-- City: string (nullable = true)
 |-- TV: double (nullable = true)
 |-- Radio: double (nullable = true)
 |-- Newspaper: double (nullable = true)
 |-- Sales: double (nullable = true)

res31: Array[org.apache.spark.sql.Row] = Array([Delhi,230.1,37.8,69.2,22.1], [Mumbai,44.5,39.3,45.1,10.4], [Kolk
ata ,17.2,45.9,69.3,12.0], [Mumbai,151.5,41.3,58.5,16.5], [Delhi,180.8,10.8,58.4,17.9])
res32: Long = 200
+----------------+-----------------+
|      Total_Sales|     Average_Sales|
+----------------+-----------------+
|3026.100000000001|15.130500000000005|
+----------------+-----------------+

+---------+-----------------+
|     City|        sum(Sales)|
+---------+-----------------+
|  Solapur|             197.7|
|  Udaipur|174.99999999999997|
|   Mumbai| 821.8000000000002|
```

```
+---------+-----------------+
|     City|        sum(Sales)|
+---------+-----------------+
|  Solapur|             197.7|
|  Udaipur|174.99999999999997|
|   Mumbai| 821.8000000000002|
|  Chennai|             186.8|
|   Jaipur|             162.1|
|   Delhi |109.50000000000001|
|Bengaluru|             206.4|
|   Nagpur|             182.9|
| Kolkata |178.39999999999998|
|    Delhi| 805.4999999999999|
+---------+-----------------+

+---------+-----------------+
|     City|     Average_Sales|
+---------+-----------------+
|Bengaluru|               17.2|
|  Solapur|16.474999999999998|
|  Chennai|15.566666666666668|
|   Nagpur|15.241666666666667|
|   Mumbai|15.218518518518522|
|    Delhi|15.198113207547168|
|  Udaipur| 14.58333333333333|
| Kolkata | 13.72307692307692|
|   Delhi |13.687500000000002|
|   Jaipur|13.508333333333333|
+---------+-----------------+
```

Applications Places System Mon Mar 30, 9:51 AM cloudera

Access documents, folders and network places cloudera@quickstart:~

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```
|     City|Max_Sales|
+---------+---------+
|  Solapur|     26.2|
|  Udaipur|     25.4|
|   Mumbai|     24.4|
|  Chennai|     21.2|
|   Jaipur|     20.2|
|   Delhi |     22.2|
|Bengaluru|     25.5|
|   Nagpur|     27.0|
| Kolkata |     21.4|
|    Delhi|     25.4|
+---------+---------+

+---------+-----+
|     City|count|
+---------+-----+
|   Mumbai|   54|
|    Delhi|   53|
| Kolkata |   13|
|  Udaipur|   12|
|  Solapur|   12|
|  Chennai|   12|
|   Jaipur|   12|
|Bengaluru|   12|
|   Nagpur|   12|
|   Delhi |    8|
+---------+-----+

+---------+------------------+
```

cloudera@quickstart:~ | cloudera@quickstart:~ | cloudera | Ad-RDD.txt (~) - gedit

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Access documents, folders and network places cloudera@quickstart:~

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```
+---------+------------------+
|     City|          Total_TV|
+---------+------------------+
|  Solapur|            1942.5|
|  Udaipur|1498.3000000000002|
|   Mumbai| 8507.300000000001|
|  Chennai|1878.8999999999999|
|   Jaipur|            1293.3|
|   Delhi |             862.3|
|Bengaluru|            2067.6|
|   Nagpur|1564.8999999999999|
| Kolkata |            1585.0|
|    Delhi|            8208.4|
+---------+------------------+

+---------+------------------+
|     City|         Avg_Radio|
+---------+------------------+
|  Solapur|21.358333333333334|
|  Udaipur|24.950000000000003|
|   Mumbai|21.281481481481478|
|   Jaipur| 28.84166666666667|
|   Delhi |23.337500000000006|
|Bengaluru| 30.39166666666667|
|   Nagpur| 30.11666666666667|
| Kolkata |21.553846153846152|
|    Delhi| 22.28867924528301|
+---------+------------------+
```

cloudera@quickstart:~ | cloudera@quickstart:~ | cloudera | Ad-RDD.txt (~) - gedit

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cloudera@quickstart:~

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```
|     City|         Avg_Sales|
+---------+------------------+
|Bengaluru|              17.2|
|  Solapur|16.474999999999998|
|  Chennai|15.566666666666668|
|   Nagpur|15.241666666666667|
|   Mumbai|15.218518518518522|
+---------+------------------+

+---------+------------------+
|     City|         Avg_Sales|
+---------+------------------+
|  Solapur|16.474999999999998|
|  Udaipur| 14.58333333333333|
|   Mumbai|15.218518518518522|
|  Chennai|15.566666666666668|
|   Jaipur|13.508333333333333|
|   Delhi |13.687500000000002|
|Bengaluru|              17.2|
|   Nagpur|15.241666666666667|
| Kolkata | 13.72307692307692|
|    Delhi|15.198113207547168|
+---------+------------------+

+--------+-----+-----+---------+-----+
|    City|   TV|Radio|Newspaper|Sales|
+--------+-----+-----+---------+-----+
|   Delhi|230.1| 37.8|     69.2| 22.1|
|  Mumbai|214.7| 24.0|      4.0| 17.4|
|Kolkata |204.1| 32.9|     46.0| 19.0|
```

cloudera@quickstart:~ | cloudera@quickstart:~ | cloudera | Ad-RDD.txt (~) - gedit

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cloudera@quickstart:~

Terminal
Use the command line

File Edit View Search Terminal

```
+--------+-----+-----+---------+-----+
|    City|   TV|Radio|Newspaper|Sales|
+--------+-----+-----+---------+-----+
|   Delhi|230.1| 37.8|     69.2| 22.1|
|  Mumbai|214.7| 24.0|      4.0| 17.4|
|Kolkata |204.1| 32.9|     46.0| 19.0|
|  Mumbai|281.4| 39.6|     55.8| 24.4|
|   Delhi|218.4| 27.7|     53.4| 18.0|
|  Mumbai|237.4|  5.1|     23.5| 17.5|
|  Mumbai|228.3| 16.9|     26.2| 20.5|
|  Mumbai|262.9|  3.5|     19.5| 17.0|
|  Mumbai|240.1| 16.7|     22.9| 20.9|
|   Delhi|248.8| 27.1|     22.9| 18.9|
|Kolkata |292.9| 28.3|     43.2| 21.4|
|  Mumbai|265.6| 20.0|      0.3| 17.4|
|  Mumbai|290.7|  4.1|      8.5| 17.8|
|   Delhi|266.9| 43.8|      5.0| 25.4|
|  Mumbai|228.0| 37.7|     32.0| 21.5|
|   Delhi|202.5| 22.3|     31.6| 16.6|
|Kolkata |293.6| 27.7|      1.8| 20.7|
|  Mumbai|206.9|  8.4|     26.4| 17.9|
|  Mumbai|239.9| 41.5|     18.5| 23.2|
|   Delhi|227.2| 15.8|     49.9| 19.8|
+--------+-----+-----+---------+-----+
only showing top 20 rows

+--------+-----+-----+---------+-----+
|    City|   TV|Radio|Newspaper|Sales|
+--------+-----+-----+---------+-----+
```

cloudera@quickstart:~ | cloudera@quickstart:~ | cloudera | Ad-RDD.txt (~) - gedit

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Change desktop appearance and behavior, get help, or log out

cloudera@quickstart:~

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```
only showing top 20 rows

+--------+-----+-----+---------+-----+
|    City|   TV|Radio|Newspaper|Sales|
+--------+-----+-----+---------+-----+
|   Delhi|230.1| 37.8|     69.2| 22.1|
|Kolkata | 17.2| 45.9|     69.3| 12.0|
|  Mumbai|151.5| 41.3|     58.5| 16.5|
|   Delhi|180.8| 10.8|     58.4| 17.9|
|  Mumbai|  8.7| 48.9|     75.0|  7.2|
|   Delhi|  8.6|  2.1|      1.0|  4.8|
|  Mumbai|199.8|  2.6|     21.2| 15.6|
|Kolkata | 66.1|  5.8|     24.2| 12.6|
|   Delhi| 23.8| 35.1|     65.9|  9.2|
|  Mumbai| 97.5|  7.6|      7.2| 13.7|
|  Mumbai|195.4| 47.7|     52.9| 22.4|
|   Delhi| 67.8| 36.6|    114.0| 12.5|
|  Mumbai|281.4| 39.6|     55.8| 24.4|
|   Delhi|218.4| 27.7|     53.4| 18.0|
|  Mumbai|237.4|  5.1|     23.5| 17.5|
|  Mumbai|262.9|  3.5|     19.5| 17.0|
|   Delhi| 97.2|  1.5|     30.0| 13.2|
|Kolkata | 95.7|  1.4|      7.4| 11.9|
|  Mumbai|290.7|  4.1|      8.5| 17.8|
|  Mumbai|206.9|  8.4|     26.4| 17.9|
+--------+-----+-----+---------+-----+
only showing top 20 rows


scala>
```

cloudera@quickstart:~ | cloudera@quickstart:~ | cloudera | Ad-RDD.txt (~) - gedit